



Finding the Kalman gain matrix

- In the previous lesson, you learned how to derive the prediction steps of the linear Kalman filter.
 - Now, we investigate the correction/update steps.
- **KF step 2a:** Estimator (Kalman) gain matrix $L_k = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}$.
 - To compute the Kalman gain, we must first compute several covariance matrices: we first find $\Sigma_{\tilde{z},k}$:

$$\begin{aligned}\tilde{z}_k &= z_k - \hat{z}_k = C_k x_k + D_k u_k + v_k - C_k \hat{x}_k^- - D_k u_k = C_k \tilde{x}_k^- + v_k \\ \Sigma_{\tilde{z},k} &= \mathbb{E}[(C_k \tilde{x}_k^- + v_k)(C_k \tilde{x}_k^- + v_k)^T] \\ &= \mathbb{E}[C_k \tilde{x}_k^- (\tilde{x}_k^-)^T C_k^T + v_k (\tilde{x}_k^-)^T C_k^T + C_k \tilde{x}_k^- v_k^T + v_k v_k^T] \\ &= C_k \Sigma_{\tilde{x},k}^- C_k^T + \Sigma_{\tilde{v}}.\end{aligned}$$

- The cross terms are zero since v_k is zero mean and is uncorrelated with $\tilde{x}_k^-$.



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- We now find $\Sigma_{\tilde{x}\tilde{z},k}^-$:

$$\begin{aligned}\mathbb{E}[\tilde{x}_k^- \tilde{z}_k^T] &= \mathbb{E}[\tilde{x}_k^- (C_k \tilde{x}_k^- + v_k)^T] = \mathbb{E}[\tilde{x}_k^- (\tilde{x}_k^-)^T C_k^T + \tilde{x}_k^- v_k^T] \\ &= \Sigma_{\tilde{x},k}^- C_k^T.\end{aligned}$$

- Combining $\Sigma_{\tilde{z},k}$ and $\Sigma_{\tilde{x}\tilde{z},k}^-$ in $L_k = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}$,

$$L_k = \Sigma_{\tilde{x},k}^- C_k^T [C_k \Sigma_{\tilde{x},k}^- C_k^T + \Sigma_{\tilde{v}}]^{-1}.$$

- **INTUITION:** Note that the computation of L_k is the most critical aspect of Kalman filtering that distinguishes it from a number of other estimation methods.
 - The whole reason for calculating covariance matrices is to be able to update L_k .
 - L_k is time-varying: It adapts to give the best update to the state estimate based on present conditions.



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- **INTUITION (continued):** Recall that we use L_k in the equation $\hat{x}_k^+ = \hat{x}_k^- + L_k(z_k - \hat{z}_k)$.
 - The first component to L_k , $\Sigma_{\tilde{x}\tilde{z},k}^-$, indicates relative need for correction to $\hat{x}_k$ and how well individual states within $\hat{x}_k$ are coupled to the measurements.
 - We see this clearly in $\Sigma_{\tilde{x}\tilde{z},k}^- = \Sigma_{\tilde{x},k}^- C_k^T$.
 - $\Sigma_{\tilde{x},k}^-$ tells us about state uncertainty at the present time, which we hope to reduce as much as possible.
 - A large entry in $\Sigma_{\tilde{x},k}^-$ means that the corresponding state is very uncertain and therefore would benefit from a large update.
 - A small entry in $\Sigma_{\tilde{x},k}^-$ means that the corresponding state is very well known already and does not need as large an update.



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- **INTUITION** (continued): Continuing to look at $\Sigma_{\tilde{x}\tilde{z},k}^- = \Sigma_{\tilde{x},k}^- C_k^T$.
 - The C_k^T term gives the coupling between state and output.
 - Entries that are zero indicate that a particular state element has no direct influence on a particular output and therefore an output prediction error should not directly update that state element.
 - Entries that are large indicate that a particular state element is highly coupled to an output so has a large contribution to any measured output prediction error; therefore, that state element would benefit from a large update.



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- **INTUITION** (continued): Still looking at $L_k = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}$,
 - $\Sigma_{\tilde{z}}$ tells us about the information quality of $\tilde{z}_k = z_k - \hat{z}_k$, where $\hat{z}_k$ is the predicted measurement, based on the present state prediction.
 - $\tilde{z}_k$ is what is unexpected or new in the measurement; we call $\tilde{z}_k$ the innovation.
 - If $\tilde{z}_k \neq 0$, we have a bad system model, state-prediction error, or sensor noise.
 - If $\Sigma_{\tilde{z}}$ is “large,” we want small, slow updates; if “small,” we want big updates.
 - This explains why we divide the Kalman gain matrix by $\Sigma_{\tilde{z}}$.
 - The form of $\Sigma_{\tilde{z}} = [C_k \Sigma_{\tilde{x},k}^- C_k^T + \Sigma_{\tilde{v}}]$ can also be explained.
 - $C_k \Sigma_{\tilde{x},k}^- C_k^T$ indicates how error in state contributes to error in output prediction.
 - $\Sigma_{\tilde{v}}$ term indicates uncertainty in sensor reading due to sensor noise.
 - Since sensor noise is assumed independent of the state, uncertainty in $\tilde{z}_k = z_k - \hat{z}_k$ adds the uncertainty in z_k to the uncertainty in $\hat{z}_k$.



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- **KF step 2b: State estimate measurement update.**
 - Computes *a posteriori* state estimate by updating *a priori* estimate using estimator gain and output prediction error $z_k - \hat{z}_k$:

$$\hat{x}_k^+ = \hat{x}_k^- + L_k(z_k - \hat{z}_k).$$
- **INTUITION:** Recall that $\tilde{z}_k = z_k - \hat{z}_k$ is the innovation; it quantifies what is unexpected or new in the measurement.
 - So, we want to use this new information to update the state, but must be careful to weight it according to the value of the information it contains.
 - L_k is the optimal blending factor between the existing information in $\hat{x}_k^-$ and the new information in $\tilde{z}_k$, as we have already discussed.



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■ KF step 2c: Estimation-error covariance measurement update.

- Finally, we update error covariance matrix:

$$\begin{aligned}\Sigma_{\tilde{x},k}^+ &= \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T = \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} \Sigma_{\tilde{z},k}^{-T} (\Sigma_{\tilde{x}\tilde{z},k}^-)^T \\ &= \Sigma_{\tilde{x},k}^- - L_k C_k \Sigma_{\tilde{x},k}^- \\ &= (I - L_k C_k) \Sigma_{\tilde{x},k}^-.\end{aligned}$$

- **INTUITION:** A covariance matrix is positive semi-definite, and $L_k \Sigma_{\tilde{z},k} L_k^T$ is also a positive-semi-definite form, and we are subtracting this from the predicted-state covariance matrix; therefore, the resulting covariance is “lower” than the pre-measurement covariance.

- Measurement update has decreased our uncertainty in state estimate.



Summary

- We have now derived the entire Kalman-filter algorithm.
- I recommend that you take some time to review the Octave implementation of the KF presented in week 4 of course 1.
 - The equations derived this week match the code implemented earlier!
- Next week, you will learn how some of these steps must be changed if the assumptions made during the derivation are violated.
- **KEY POINT OF THIS DERIVATION:** Repeating from before, recall that the estimator output comprises the state estimate $\hat{x}_k^+$ and error covariance estimate $\Sigma_{\tilde{x},k}^+$.
 - That is, we have high confidence that the truth lies within $\hat{x}_k^+ \pm 3\sqrt{\text{diag}(\Sigma_{\tilde{x},k}^+)}$.
- **COMMENT:** If a measurement is missed for some reason, then skip steps 2a–c for that iteration. That is, set $L_k = 0$ and $\hat{x}_k^+ = \hat{x}_k^-$ and $\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^-$.